Manual of Operation and Instruction

Troxler RoadReader™

Model 3430 Surface Moisture-Density Gauge



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PN 110990
February 2021
Edition 1.3

⚠ SAFETY ALERT SYMBOL ⚠

The Safety Alert Symbol shall appear within this manual. Wherever it appears in this manual or on safety signs affixed to the machine, this is to make all aware of the potential for personal injury and to be cautious when these images are present.

Always observe all WARNING, CAUTION, and NOTE recommendations listed within this manual before operating the machine.







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ABOUT THIS MANUAL

The Model 3430 *Manual of Operation and Instruction* provides detailed information about this gauge. The manual includes product safety information, as well as instructions for the proper installation and use of the Model 3430 gauge.

This manual is organized as follows:

Chapter 1, Introduction – Provides information on the safe use of the gauge; a brief overview of the unit and its features; a list of parts and accessories; and instructions for unpacking and inspection.

Chapter 2, Theory of Operation – Provides a brief description of how the gauge's operations work in relation to their sources.

Chapter 3, Using the Gauge – Describes the keypad and provides instructions for setting up, starting, and operating the gauge.

Chapter 4, Advanced Gauge Operation –Describes the options available from the gauge's Special menu.

Chapter 5, Special Functions – Describes the gauge's special functions, such as stat and drift tests and memory reset.

Appendix A, Maintenance & Troubleshooting – Provides maintenance and service information, as well as instructions for basic troubleshooting.

Appendix B, Specifications – Contains mechanical, electrical, and environmental performance specifications.

Appendix C, Transporting and Shipping – Provides information about shipping requirements for the United States and Canada.

Appendix D, Radiation Safety & Theory – Provides a radiation primer, regulatory requirements, and gauge safety precautions.

Appendix E, Standard Count Log – Use this form to record your standard count readings.

Appendix F, Unit Conversion – Provides a conversion for measurement and radiological units.

HOW TO USE THIS MANUAL

Congratulations on the purchase of the Troxler Model 3430 Surface Moisture-Density Gauge.

The Model 3430 *Manual of Operation and Instruction* contains information on how the Model 3430 operates and provides directions on the use of this gauge. Site selection, basic parameter setup, moisture and density determination, data storage, and advanced operations are included, along with radiological information and system troubleshooting.

CONVENTIONS USED IN THIS MANUAL

Throughout this manual the following symbols and special formatting are used to reveal the purpose of the text.



⚠ WARNING

Warnings indicate conditions or procedures that, if not followed correctly, may cause personal injury.

⚠ CAUTION

Cautions indicate conditions or procedures that, if not followed correctly, may cause equipment damage.

NOTE

Notes indicate important information that must be read to ensure proper operation.

⟨**KEY**⟩ This style indicates a key or character to press on the ADU keypad.

DISPLAY–Typestyle and shading used to simulate the control panel display

1. Indicates a procedure with multiple steps.

♦ Indicates a list of things needed (such as equipment) or important points to know.

▶ Indicates that more than one option is available. Carefully select the option that applies.

CAUTIONS AND WARNINGS



Units intended for use in countries that are members of the European Community are shipped with an AC adapter, Troxler part number 108354.



Gauge cover is to be removed by trained service personnel only. There are no user-serviceable components inside. Note that components behind the cover can have voltage potentials in excess of 50 volts during normal operation of the gauge.



Appendix D, Radiation Theory and Safety should be read carefully and understood before using the gauge.



The source rod should automatically retract to the **SAFE** position when the gauge is lifted by the handle.



The Model 3430 gauges are not waterproof. Please do not use them in the rain. If the gauge gets wet, make sure it is completely dry before sealing it in the case for storage. Even small amounts of moisture can get into the body of the gauge and cause damage. If you suspect that moisture may be inside the gauge open the keypad panel and run a fan or hairdryer (on low) in the gauge for an hour or more until dry.

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WARRANTY

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The Leader in Construction Testing Equipment

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ATTENTION MODEL 3430 GAUGE OWNER

This unit contains functions that require an Access Code to be entered. This allows some control over the access to these functions. If you would like management to retain this control, remove this page upon receipt of the gauge and file it somewhere safe.

THE ACCESS CODE IS FOR THIS GAUGE

IS: 4678

NOTES

Chapter 1: Introduction

This chapter covers the following topics and tasks:

- ✓ An introduction to your new Model 3430 gauge
- ✓ Inspecting and unpacking
- ✓ Included parts and accessories

Introduction

The Model 3430 Surface Moisture-Density Gauge can quickly and precisely determine the moisture and density of soils, soil bases, aggregate, concrete and asphaltic concrete without the use of core samples or other destructive methods.

Using direct transmission or backscattered gamma radiation, the Model 3430 gauge determines the density of materials by counting the number of photons emitted by a cesium-137 source. Geiger-Mueller (G-M) detectors located in the gauge base detect the gamma radiation and a microprocessor converts the counts into a density reading.

Using the principle of neutron thermalization, the Model 3430 determines the moisture content of soils and soil-like materials. Hydrogen (water) in the material slows neutrons emitted from an americium-241: beryllium source. Helium-3 detectors located in the gauge base detect the slowed neutrons.

The nuclear method of testing density and moisture has been approved by the American Society of Testing and Materials (ASTM). The Model 3430 meets or exceeds all applicable American Society of Testing and Materials (ASTM) standards (or corresponding equivalent), including:

- ASTM C–1040: Standard Test Methods for In-Place Density of Unhardened and Hardened Concrete, Including Roller Compacted Concrete, By Nuclear Methods
- ASTM D-2950: Standard Test Method for Density of Bituminous Concrete in Place by Nuclear Method.
- ASTM D-6938: Standard Test Methods for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)

Some information contained in this manual is used in training courses offered by Troxler Electronic Laboratories, Inc. and to assist purchasers in obtaining a Radioactive Materials License from the U.S. Nuclear Regulatory Commission or an Agreement State. Owners of this gauge must maintain a current radioactive materials license as long as they own the gauge, even if it is in storage and not actively being used.

Any licensing issues discussed in this manual are for the United States. To purchase a Model 3430 in Canada, owners must obtain a radioisotope license from the Canadian Nuclear Safety Commission (CNSC). The owner should obtain copies of the CNSC Regulations and the Transportation of Dangerous Goods Act and Regulations. This manual provides a guide to Canadian shipping requirements in Appendix D.

Owners are encouraged to require study of this manual by users before allowing any use of the instrument. To monitor exposure to radiation, personnel should wear a dosimeter while operating or cleaning the gauge. The sections of the manual covering radiation safety should be required reading for all operators and potential operators. **If these sections are not completely understood, users should seek assistance from Troxler, an appointed Troxler representative or others designated within the user organization.** Additional radiation safety information is available by attending a *Troxler Nuclear Gauge Training Course*.

As changes are made to local, state, and federal regulations on a continuing basis, the owner/user must maintain knowledge of these regulations**.** *The responsibility for compliance ultimately falls upon the owner***.** The owner may also wish to purchase and subscribe to Titles 10 and 49 of the *Code of Federal Regulations* in addition to applicable local/state regulations.

Gauge Parts and Accessories

Use Figure 1 and the list below to identify the gauge and parts as they are unpacked.

- The **Gauge** is the portable instrument containing all electronic modules, the rechargeable battery pack, detectors, and the radioactive sources.
- The **Reference Standard Block** provides a measurement standard for standard counts and is used during stat and drift tests.
- The **Drill Rod** is used to drill holes for direct transmission measurements. ***Do not use the source rod of the gauge to drill holes.***
- The **Scraper Plate / Drill Rod Guide** is used to prepare the test site and to guide the drill rod when preparing the site for direct transmission measurements.
- The **Extraction Tool** provides leverage to remove the drill rod from soil materials.
- The **AC Charger** and **DC Adapter** are used to charge the gauge batteries. The AC charger accepts 90 – 220 V AC, 50/60 Hz and supplies 12 V DC. The DC adapter allows recharging from an automobile accessory power outlet.
- The **Transport Case** provided with the gauge has been approved as a Type A package and should not be altered. Always use this transport case when transporting or shipping the gauge.
- The ***Manual of Operation and Instruction*** details how to use the gauge.

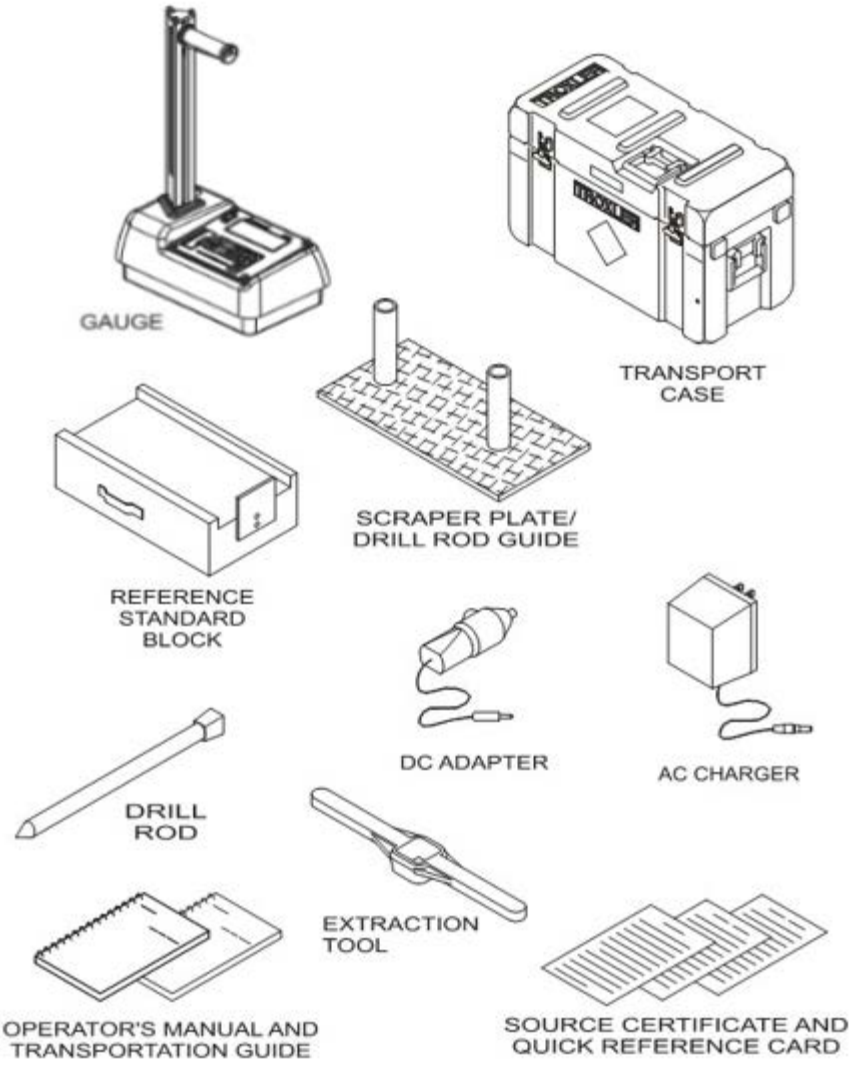


Figure 1. Model 3430 Gauge and Accessories

Unpacking and Inspection

Troxler recommends that all operators wear a dosimeter while working with the gauge. Upon receipt of the gauge from the factory, perform a complete inspection and inventory. If the shipping case and/or any other part or accessory appears damaged, notify the carrier and your Troxler Representative immediately.

Save the box and any packing material for shipping to another location or back to the factory

Check the shipping case for the following:

- Gauge
- Reference Standard Block
- Drill Rod
- Scraper Plate/Drill Rod Guide
- Extraction Tool
- AC Charger
- DC Adapter (for a vehicle cigarette lighter)
- *Manual of Operation and Instruction*
- Gauge Warranty
- Source Certificate
- *Transportation Guide* (This guide refers to U.S. standards. All other countries please refer to local regulations. In the absence of local regulations, please use this guide as a reference only.)

⚠ WARNING

The source rod should always be locked in the SAFE position when the gauge is not in use.

NOTE

Charge the batteries for three hours prior to initial use.

Complete the unpacking and inspection by following these steps:

1. Lift the gauge from the transport case and inspect the outside surface for damage.
2. Check the lock on the source rod handle and make sure the keys fit.
3. Remove the lock, release the trigger, and check the source rod operation. It should move up and down with minimal effort.
4. Replace the handle lock and return the gauge to the transport case.

NOTES

Chapter 2:
Theory of Operation

This chapter covers the following topics and tasks:

- ✓ Theory of operation and modes of transmission
- ✓ Overview of density and moisture measurements

Density

The Model 3430 gauge utilizes two modes of operation: *direct transmission mode* (source rod extended into the material) and *backscatter mode*. Figure 2 and Figure 3 9++6illustrate these two modes of operation.

In *direct transmission* mode, the rod containing the Cesium-137 (8 mCi/0.3 GBq) source is lowered to the desired depth. The detectors in the gauge base measure the radiation emitted by the source rod. Gamma photons reaching the detectors must first pass through the material, colliding with electrons present in the material. Generally, the lower the number of photons that reach the detectors, the denser the material is.

In *backscatter* mode, the gamma photons that enter the material must be scattered (or reflected) to reach the detectors. With the rod locked in the first notch, the source and detectors are in the same plane, referred to as the backscatter position. Photons emitted from the source penetrate the material and the detectors measure the scattered photons.

While the direct transmission geometry measures the average density of the material from the source to the surface, the backscatter geometry yields an average heavily weighted by the density close to the surface.

Figure 4 shows two normalized *top layer effect curves*, illustrating the percentages of photons at the detectors for various depths. The two curves can be used to compute the gauge response to layered material of different densities. For example, the density of the top inch of a surface layer accounts for about 52% of the backscatter density measurement.

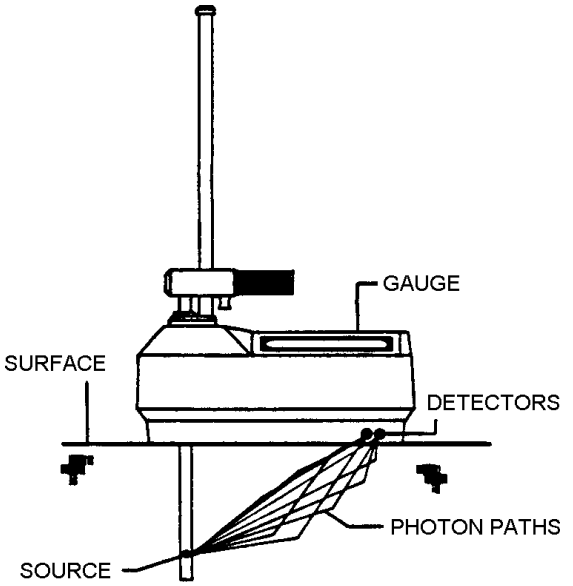


Figure 2. Direct Transmission Geometry

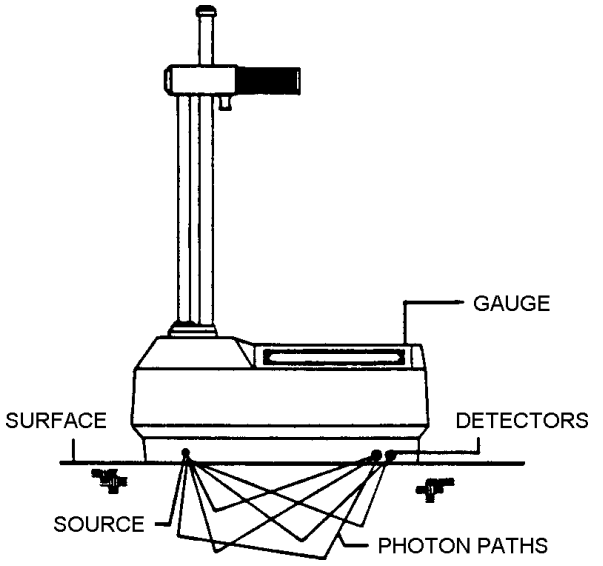


Figure 3. Backscatter Geometry